

AI Operations Agent

A LangGraph Multi-Tool Assistant to Eliminate Manual Cross-System Busywork

Business Case & Problem Statement

Prepared for: Meridian Retail Group — Global Operations & Analytics

Prepared by: AI Automation Initiative

Status: Approved for Pilot Build

1. Executive Summary

Meridian Retail Group's operations team currently spends the majority of its analyst hours manually pulling information from a dozen disconnected public and internal sources — shipping-route weather, competitor pricing, supplier country/currency data, candidate screening, crypto/FX exposure, industry news, and more — and then hand-assembling the results into PDF, Excel, and Word reports for leadership. This is slow, error-prone, and expensive. This document lays out the problem in concrete cost terms and proposes a single conversational **AI Operations Agent**, built on LangGraph, that can call all of these data sources as tools, generate the final documents automatically, and pause for explicit human approval before doing anything irreversible (like deleting a file).

Bottom line: the current process costs Meridian an estimated **\$186,800/year** in analyst time and delay-related losses. The proposed agent is estimated to cut that by roughly **70–80%** within one quarter of rollout.

2. The Problem

Every week, a team of four operations analysts repeats the same manual workflow across unrelated websites and spreadsheets to answer questions that are individually simple but collectively expensive to answer by hand:

Task	Manual Source Today	Hours / Week	Pain Point
Check weather at 15 shipping hubs before routing decisions	Manual browser lookups	4.5	Delayed routing, occasional missed weather-driven delays
Competitor price checks on ~40 SKUs	Manual store browsing	6.0	Stale prices by the time a decision is made
Supplier country / currency / holiday checks	Manual search per country	3.5	Missed local holidays causing shipment delays
Candidate sourcing/screening summaries for HR	Manual resume review	5.0	Inconsistent shortlist quality
Crypto/FX exposure snapshot for finance	Manual exchange lookups	2.0	Numbers already stale in the finance meeting
Industry/tech news roundup	Manual browsing of forums/news	3.0	Inconsistent coverage, analyst bias
Assembling weekly PDF/Excel/Word reports	Manual copy-paste + formatting	8.0	~1 full day per analyst just on formatting
TOTAL (per analyst, weekly)		32.0	

At 4 analysts × 32 hours/week × 48 working weeks × a fully-loaded rate of \$38/hour, this is **≈\$233,472/year** in raw labor time alone before counting the cost of delayed or lower-quality decisions caused by stale data (estimated conservatively at \$15,000–\$20,000/quarter in missed pricing and shipping-delay avoidance).

3. Business Impact if Left Unaddressed

- Analyst time is spent on repetitive lookups instead of analysis — a direct opportunity cost.
- Report turnaround (currently 1–2 days) delays pricing, routing, and hiring decisions.
- Manual copy-paste between browser tabs and Office documents is a recurring source of errors.
- No consistent audit trail of what data was pulled, from where, or when.
- Scaling to more product lines or regions means linearly scaling headcount, not automation.

4. Proposed Solution: The AI Operations Agent

We propose a single conversational agent, built with **LangGraph**, that exposes every data source above as a callable tool behind one chat interface. An operator (or another internal system) simply asks in plain language — e.g. *"Check the weather at our Rotterdam and Singapore hubs, pull the current EUR/USD and BTC/USD rates, and put it all in a one-page PDF for the 9am call"* — and the agent plans, calls the right tools in the right order, and produces the finished document.

4.1 Tool Catalog

All data-source tools below are free, public APIs requiring no paid key (one, CoinGecko, is free for this volume of use but may rate-limit at high frequency). Each is wired into the agent as a distinct LangGraph tool.

Domain	Tool	Provider / Link	What it does
Weather & Logistics	get_weather	open-meteo.com	Current weather & forecast for any lat/long — used for shipping-route risk checks
E-Commerce	search_products	dummyjson.com	Product search/filter/pricing lookups for competitor-style checks
Trade & Travel	get_country_info	restcountries.com	Country profile: currency, timezone, region, borders
HR	get_candidates	randomuser.me	Generates structured candidate-style profiles for sourcing/screening demos
Public Holidays	get_public_holidays	date.nager.at	Official public holidays by country/year — for shipping/HR planning
Finance	get_crypto_price	coingecko.com	Live crypto prices in USD/EUR/etc.
Finance	convert_currency	open.er-api.com	Live FX conversion rates
Geography	geocode_location	nominatim.openstreetmap.org	Turns a place name into coordinates for the weather tool
News	get_hn_top_stories	Hacker News API (Firebase)	Top tech/industry stories right now
Knowledge	search_wikipedia	en.wikipedia.org REST API	Quick factual lookups / summaries
Utilities	calculate	api.mathjs.org (+ local fallback)	Safe arithmetic / unit conversion
Utilities	test_http_request	httpbin.org	Sanity-checks outbound HTTP calls / integrations
Entertainment	search_tv_show	tvmaze.com	Show/cast/schedule lookups (team offsite planning, etc.)
Team Culture	tell_joke	v2.jokeapi.dev	A programming/pun joke — for standup icebreakers
Team Culture	get_advice	api.adviceslip.com	Random one-line advice — lighter touch for reports/standups
Team Culture	get_cat_fact	catfact.ninja	A random cat fact — same purpose
HR / Fun	predict_age	api.agify.io	Predicts an age range from a first name (demographic estimation demo)

Domain	Tool	Provider / Link	What it does
Demo / Games	get_pokemon_info	pokeapi.co	Included as a lightweight, zero-risk demo tool for training/testing
Demo	get_latest_launch	api.spacexdata.com	Latest SpaceX launch info — used to demo event/status-style lookups
Demo	get_trivia_question	opentdb.com	Generates a trivia question — used to test structured-output handling

4.2 Internal Tools (no external API)

Tool	What it does
create_file / write_file / read_file / list_files	Manage working files the agent produces (drafts, data pulls, logs)
delete_file	Deletes a file — gated behind mandatory Human-in-the-Loop (HITL) approval, see 4.3
generate_pdf_report	Builds a formatted PDF (ReportLab) — the same technique used to produce this document
generate_excel_report	Builds a formatted .xlsx workbook (openpyxl) for finance/ops data
generate_word_doc	Builds a formatted .docx (python-docx) for narrative reports/memos

4.3 Human-in-the-Loop Safety

The agent can read, write, and generate files autonomously, but **file deletion always pauses the workflow** and asks a human to explicitly confirm before anything is removed. This is implemented as a genuine LangGraph interrupt (not just a print statement) — the graph checkpoints its state, halts before the delete executes, and only resumes once a human replies. No irreversible action ships without a person in the loop.

5. How It Works (Architecture)

- 1. Operator sends a request in plain English to the agent.
- 2. The Agent node (gpt-4o-mini) decides which tool(s), if any, are needed.
- 3. The Tools node executes the chosen tool(s) — weather, pricing, FX, file ops, document generation, etc.
- 4. Control returns to the Agent node, which decides whether more tool calls are needed or the task is done.
- 5. If the task involves delete_file, the graph interrupts and waits for explicit human approval before continuing.
- 6. The Agent returns a final answer and/or a generated PDF / Excel / Word file.

The reference implementation (companion notebook: *langgraph_ops_agent.ipynb*) runs directly in Google Colab, loading API keys from Colab Secrets, and includes a one-cell health check that pings every external API listed in section 4.1 so the operator can confirm live availability before a demo.

6. Expected ROI

Metric	Today (Manual)	With AI Operations Agent	Impact
Analyst hours / week on lookups + formatting	32 hrs × 4 analysts = 128 hrs	~25–35 hrs total	≈75% time recovered
Annual labor cost of this workflow	≈\$233,000	≈\$45,000–\$60,000	≈\$175,000–\$190,000 saved/year
Report turnaround time	1–2 days	Minutes	Same-day decisions
Data freshness at time of decision	Hours to days old	Live at request time	Fewer stale-data errors
Destructive-action safety	Ad hoc / manual care	Mandatory human approval on delete	Lower operational risk

7. Implementation Roadmap

Phase	Duration	Deliverable
1. Pilot build	Week 1–2	Notebook agent covering weather, pricing, FX, and PDF/Excel/Word generation
2. HITL & file safety hardening	Week 3	LangGraph interrupt-based approval flow for all destructive actions
3. Team pilot (1 ops squad)	Week 4–6	Daily use on real reporting tasks; collect time-saved metrics

Phase	Duration	Deliverable
4. Rollout	Week 7–10	Roll out to all ops + HR + finance analysts; add team-specific tools as needed

Recommendation: Approve the pilot build described above. All required data sources are free, public APIs — the only recurring cost is LLM usage (gpt-4o-mini), estimated at well under \$50/month for this team's volume.

Appendix A: Full API Reference

Live links for every external API used by the companion notebook. All are free for the request volumes described in this document; a few (marked *) may apply light rate-limiting under heavy use.

API	Base URL	Auth Required
Open-Meteo	https://api.open-meteo.com/v1/forecast	No
DummyJSON	https://dummyjson.com/products/search	No
REST Countries	https://restcountries.com/v3.1	No
RandomUser	https://randomuser.me/api/	No
Nager.Date (holidays)	https://date.nager.at/api/v3/PublicHolidays	No
CoinGecko*	https://api.coingecko.com/api/v3/simple/price	No
Open Exchange Rates (open.er-api.com)	https://open.er-api.com/v6/latest	No
Nominatim (OpenStreetMap)*	https://nominatim.openstreetmap.org/search	No (custom User-Agent required)
Hacker News (Firebase)	https://hacker-news.firebaseio.com/v0/	No
Wikipedia REST API	https://en.wikipedia.org/api/rest_v1/page/summary/	No
Math.js API*	https://api.mathjs.org/v4/	No
HTTPBin	https://httpbin.org/	No
TVMaze	https://api.tvmaze.com/search/shows	No
JokeAPI	https://v2.jokeapi.dev/joke/Any	No
Advice Slip	https://api.adiceslip.com/advice	No
Cat Facts	https://catfact.ninja/fact	No
Agify	https://api.agify.io/	No
PokeAPI	https://pokeapi.co/api/v2/pokemon/	No
SpaceX API (community)*	https://api.spacexdata.com/v4/launches/latest	No
Open Trivia DB	https://opentdb.com/api.php	No

Note: public free APIs occasionally change, rate-limit, or have downtime outside anyone's control. The companion notebook includes a health-check cell that calls each endpoint and reports OK/FAIL at run time, so availability can always be confirmed live rather than assumed from this document.